# Kareem Taleb-Agha

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## Education

**University of Michigan-Dearborn** Dearborn, MI
*Bachelor of Science in Computer Science* *2024*

- Relevant Coursework: Linear Algebra, Data Structures & Algorithms, Operating Systems, Intro to Artificial Intelligence, Computer Networking, Computers & Security, Web Technology & Mobile Development

## Experience

**Ford Motor Company** January 2025 – Present
*Software Engineer I — Spring Boot, Kubernetes, TypeScript, FastAPI, NextJS* *Dearborn, MI*

- Built data parsers for a high-throughput cloud ingestion platform, decoding and normalizing streaming messages for downstream analytics
- Helped design and build a self-service portal (FastAPI, Next.js) that gives teams access to an internal streaming data feed
- Built backend APIs for a data validation tool that cross-checks Postgres and MongoDB for inconsistent data
- Took ownership of an internal platform in a large application suite; built cross-app data synchronization so users can reach analytics without switching apps
- Built an LLM-powered feature for an internal AI assistant that summarizes end-of-shift reports into a single overview
- Delivered a zero-downtime authentication update supporting two identity providers across a fleet of Spring Boot microservices
- Piloted OpenCode, an open-source agentic AI coding tool, which leadership then adopted more widely
- Directed an AI coding agent to build a replay service that recovered data after an ingestion outage
- Cut database costs by adding data-retention policies across many MongoDB databases and decommissioning unused ones

**Hexagon Manufacturing Intelligence** May 2023 – July 2023
*Software Engineering Intern — Python* *Novi, MI*

- Decreased runtime of automated ADAMS (multi-body dynamics simulation software) regression runs by over 80% by refactoring the simulation runner to parallelize workloads via Python's multiprocessing library, leveraging multi-core server hardware.

## Projects

**Agentic Development Harness** | *Claude Code, Next.js, TypeScript, Playwright* October 2026

- Built a multi-agent harness on Claude Code that develops a production Next.js site, pairing a coding agent with an independent reviewer agent that iterate through multi-round feedback until review passes
- Designed guardrails and human-in-the-loop controls: a spec-driven CLAUDE.md with hard constraints (no unapproved publishing, no fabricated facts, no secrets), lint/build/review gates before every commit, and owner-approval checkpoints
- Wrote Playwright tooling for the reviewer agent to visually verify pages across screen sizes and themes; packaged repeatable workflows as agent skills that turn plain-language requests into validated commits

## Technical Skills

**Languages**: Java, Python, C/C++, Postgres, JavaScript, TypeScript, Golang
**Frameworks**: Spring Boot, FastAPI, React, Node.js, Angular, JUnit, NextJS
**Developer Tools**: Git, Docker, Kubernetes, GCP, Tekton, Jira, MongoDB, BigQuery, Azure, Dynatrace